

$$\lim_{n \rightarrow \infty} X_n$$

$$\omega \in \Omega \quad X_n(\omega)$$

X_n è la frequenza relativa

$$\omega: \Omega_1 \quad X_n(\omega) \xrightarrow{n \rightarrow \infty} p$$

evento $\Omega_1 \in \Omega$

$$P[\Omega_1] = 1 \Leftrightarrow P[\Omega - \Omega_1] = 0$$

$$E[|X_n - p|^2] \xrightarrow{n \rightarrow \infty} 0$$

converge in media quadratica

$$\begin{aligned} &\downarrow \\ &\{E[X_n] = p\} \quad \text{var}[X_n] \\ &\downarrow \quad \downarrow \\ &0 \quad 0 \end{aligned}$$

Convergenza in prob.
 $\forall \varepsilon > 0$

$$P[|X_n - p| > \varepsilon] \xrightarrow{n \rightarrow \infty} 0$$

$$X_n - p = \begin{cases} 0 & \text{prob. } 1 - \frac{1}{n} \\ n & \text{prob. } \frac{1}{n} \end{cases}$$

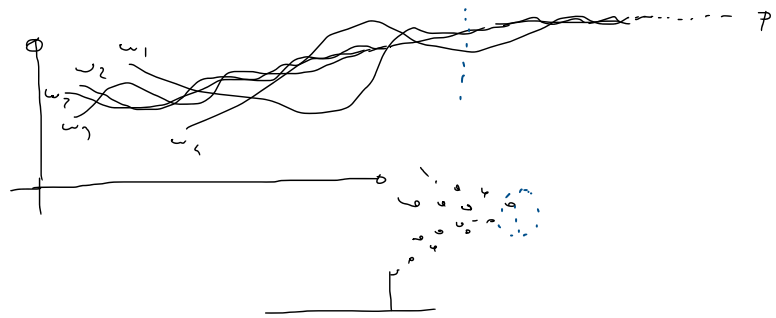
non troppo grande

$$\forall \varepsilon > 0 \quad P[|X_n - p| > \varepsilon] = \frac{1}{n}$$

$$P[|X_n - p| > \varepsilon]$$

$$E[|X_n - p|^2] = 0^2 \times (1 - \frac{1}{n}) + n^2 \frac{1}{n} = n$$

$$P[K > 3] = \frac{1}{4}$$

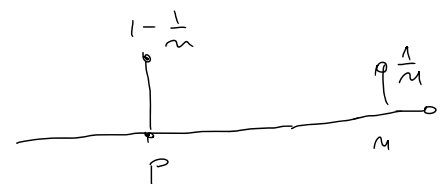


conv. q.o.



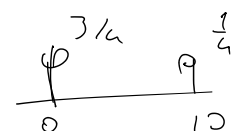
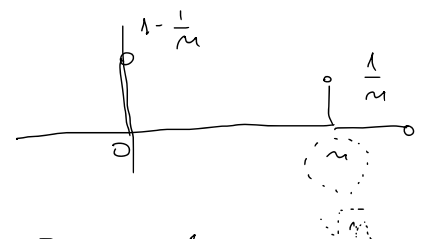
conv. a.s. $\Leftrightarrow$ conv a.e.

PMF di X_n

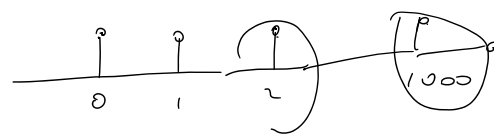
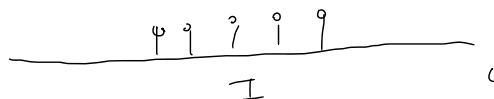


$$X_n = \begin{cases} p & \text{prob. } 1 - \frac{1}{n} \\ p+n & \text{prob. } \frac{1}{n} \end{cases}$$

PMF $X_n - p$



$$P[X \in I] = \sum_{i \in I} P_i$$



$$\begin{cases} X=0 \\ X=1 \end{cases}$$

Chebichev

$$\begin{cases} X=2 \\ X=4 \end{cases} \rightarrow$$

$$P\left[|X_n - E[X_n]| > k \sigma_{X_n}\right] < \frac{1}{k^2} = \frac{\sigma_{X_n}^2}{\varepsilon^2} \rightarrow 0$$

$\downarrow$

P

$$k \sigma_{X_n} = \varepsilon$$

$$k = \frac{\varepsilon}{\sigma_{X_n}}$$

$$(X_n - E[X_n])^2 = [(X_n - P) + (\Delta_n)]^2$$

$$\Delta_n = P - E[X_n]$$

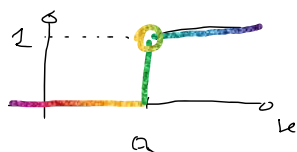
$$\begin{aligned} \sigma_{X_n}^2 &= E[(X_n - P) + \Delta_n]^2 = E[(X_n - P)^2] + \Delta_n^2 + 2\Delta_n E[X_n - P] \\ &= E[(X_n - P)^2] + \Delta_n^2 + 2\Delta_n \underbrace{E[X_n] - P}_{-\Delta_n} \\ &= E[(X_n - P)^2] + \Delta_n^2 - 2\Delta_n^2 = E[(X_n - P)^2] - \Delta_n^2 \end{aligned}$$

$$E[(X_n - P)^2] = \Delta_n^2 + \sigma_{X_n}^2$$

$$E[(X_n - P)^2] \rightarrow 0 \Leftrightarrow \begin{cases} \Delta_n \rightarrow 0 \\ \sigma_{X_n}^2 \rightarrow 0 \end{cases} \quad \begin{cases} P[X_n < P] \xrightarrow{n \rightarrow \infty} 0 \\ P[X_n > P] \xrightarrow{n \rightarrow \infty} 1 \end{cases}$$

$$X_n \rightarrow F_{X_n}(x) \quad \text{LA CDF di } X_n$$

$$F_{X_n}(x) \xrightarrow{n \rightarrow \infty} F_X(x)$$



$$X = a \vee \sim$$

$$F_X(x) = P[X \leq x] = \begin{cases} 0 & x < a \\ 1 & x \geq a \end{cases}$$

$$a_1, a_3, a_1, a_5, a_2, a_1, a_2$$

1 -

1 -

1 -

1 -

1 -

1 -

1 -

$$|-3+4| < |-3| + |4|$$

$$|\bar{x}_n - \chi| = \left| \sum_i a_i (f_n(e_i) - p_x(e_i)) \right| \leq \sum_i |a_i| (f_n(e_i) - p_x(e_i)) = \sum_i |a_i| |f_n(e_i) - p_x(e_i)|$$

$$N_i \in \{0, 1, 2, \dots, \frac{n}{\phi}\}$$

$$P[U_i=k] = \binom{n}{k} p_k^k (1-p_k)^{n-k}$$

$$E[N_0^2] = \sum k^2 P[N_i = k]$$

$$\frac{N_i}{n} =$$

$$P[N_i = n | P_X(a_i)] \xrightarrow{n \rightarrow \infty} 1$$

$$q_x(a_i) \quad P\{N_i = n \mid q_x(a_i)\} \xrightarrow{n \rightarrow \infty} 0$$

$$\frac{N_c}{n} \Rightarrow \frac{1}{3} \quad N_c \sim \frac{n}{3}$$

$\rho(i) \mid N_i \sim q(u_i)$

$$9 \neq \frac{1}{3}$$

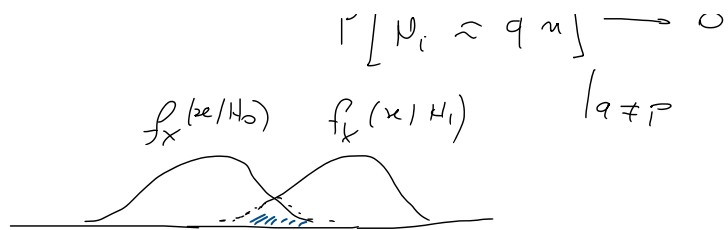
$$f(p) = p \log \frac{1}{p} + (1-p) \log \frac{1}{1-p} = -p \log p - (1-p) \log (1-p)$$

$$\frac{N_i}{n} \rightarrow P$$

$$P[N_i \approx P_n] \xrightarrow{n \rightarrow \infty} 1$$

$$P[\mu_i \approx qn] \xrightarrow{n \rightarrow \infty} 0$$

$$O(\ln n) \quad O(n \ln n) \quad |_{a \rightarrow \infty}$$



H_0

Falso allora se decido per 2 questo è vero

H_1

Vero allora se decido per 0 questo è vero
- - - - -

Probabilità di errore è la prob. di decidere per 1
questo è vero è 1.